Anti-parasitic Drugs

Mechanisms, Clinical Use, and Drug Selection

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Learning Objectives

1. Classify antiparasitic drugs based on parasite type
2. Understand key mechanisms of action
3. Select appropriate therapy for common infections
4. Recognize major adverse effects and limitations
5. Apply clinical patterns to select appropriate therapy

البروتوزوا

الكائنات وحيدة الخلية

طرق انتقال الديدان الطفيلية!!

HELMINTHS

الديدان الطفيلية

Ectoparasites

الطفيليات الخارجية

مثل القمل

How Do We Approach Antiparasitic Therapy?

Effective treatment requires:

- Identification of the type of parasite
- Consideration of site of infection (luminal vs tissue)
- Mechanism of action guides drug selection

(Location of parasite (e.g., luminal vs tissue) determines drug choice)

Antiparasitic drugs are **organism-specific**, unlike broad-spectrum antibiotics

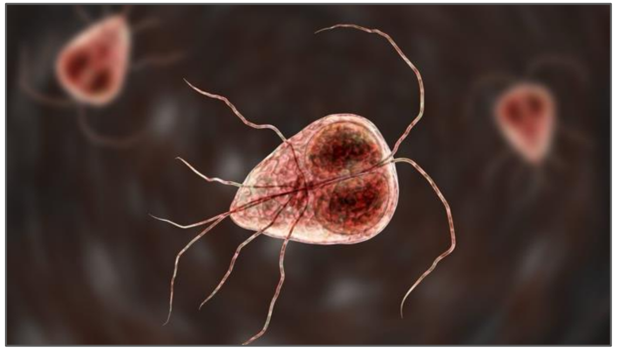
Classification

Category	Organisms	Key Drugs
Protozoa	Giardia, Entamoeba, Plasmodium	Metronidazole, Artemisinin
Helminths	Nematodes, Cestodes, Trematodes	Albendazole, Ivermectin, Praziquantel
Ectoparasites	Scabies, lice	Permethrin, Ivermectin

PROTOZOA

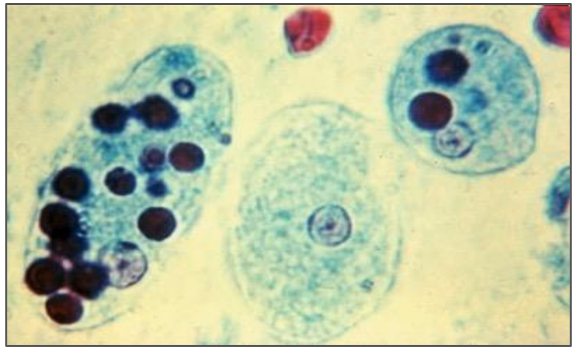
الأوليات

Unicellular organisms causing intestinal and systemic infections



Giardia
(jee-AR-dee-uh)

Giardiasis
(jee-ar-DYE-uh-sis)



Entamoeba
(en-tuh-MEE-buh)

Amoebiasis
(am-uh-BYE-uh-sis)

Plasmodium
(plaz·mow·dee·uhm)

Malaria
(muh-LAIR-ee-uh)

Protozoal Infections

- Unicellular organisms
- **Infections:**
  - Gastrointestinal tract (Giardia, Entamoeba)
  - Blood (Plasmodium)

- **Treatment depends on:**

  Luminal vs. tissue (invasive) infection

  - Luminal drugs: eradicate parasites in the gut
  - Tissue drugs: treat invasive disease
  - For some infections, you need both drugs:

    Tissue drug (Metronidazole) + luminal drug (e.g., Paromomycin)

Type	Definition	Treatment
Luminal	Parasite in intestine only	Luminal drugs
Tissue	Parasite invades tissues	Systemic drugs

Metronidazole

Mechanism of Action (MOA):

- A prodrug, gets activated in anaerobic organisms
- Reduced in anaerobic organisms → forms free radicals

  → DNA damage → cell death
- Does **NOT** reliably eradicate luminal cysts

  May require luminal agent

Pharmacokinetics:

- Excellent tissue penetration (treats invasive disease)
- Available as oral and IV

A treatment for:

- Giardia. - Entamoeba. - Anaerobic bacteria

Clinical Use:

1. Giardiasis
2. Amoebiasis (tissue infection)
3. Often combined with luminal agents

Adverse Effects:

1. Metallic taste
2. Nausea, vomiting
3. Disulfiram-like reaction with alcohol
4. Peripheral neuropathy (prolonged use)

Antimalarial Drugs

Classification:

Artemisinin (Artemether), Quinolines (Chloroquine), Antifolates (Pyrimethamine)

- Resistance patterns influence drug choice

Class	Artemisinin	Quinolines	Antifolates
MOA	**Free radical formation:** Activated by iron → Produces free radicals → Damages parasite proteins and membranes	**Inhibits heme detoxification:** ▪ Inhibits conversion of toxic heme → hemozoin (non-toxic) ▪ Leads to parasite toxicity	**Inhibits folate synthesis:** Inhibiting the folate biosynthetic pathway in the parasite (essential for DNA synthesis and cell replication)
Use	First-line for Plasmodium falciparum	Ranges from treating active blood-stage infections to preventing relapses and transmission	Prophylaxis (in pregnancy, infants, travelers)
S.E	Generally, well tolerated	Retinopathy (long-term)	▪ Bone marrow suppression ▪ High-dose folic acid supplementation can reduce its efficiency

IMPORTANT

Greasy = **Giardia**
Bloody = Entamoe**ba**
Cyclic fever = Malaria

Pattern Recognition (PROTOZOA)

Table is for illustration (know what is important)

Feature	**Giardia**	**Entamoeba**	**Malaria**
Stool	**Greasy (foul-smelling)**	**Bloody**	None
Fever	No	Mild	High
Location	Intestine*	Intestine + tissue	Blood (RBC)
Key symptom	**Bloating**	**Amoebic Dysentery**	**Cyclic fever**
Medication	Metronidazole	Metronidazole + luminal agent	Artemisinin-based combination therapy (ACT)

***Giardia:** attached to the intestine mucosal surface (not floating), which helps systemically absorbed metronidazole work effectively
Note: blood → think invasion, not just luminal infection

HELMINTHS

الديدان الطفيلية

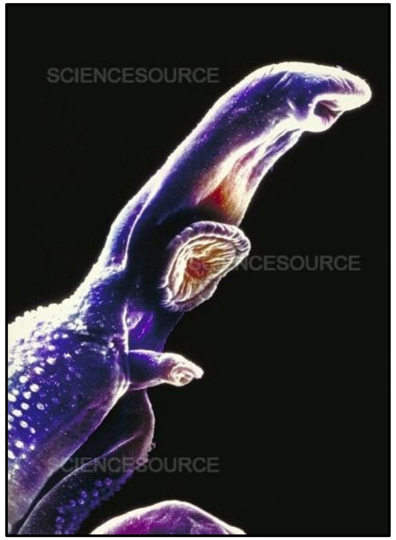
Multicellular parasites affecting GI tract and tissues

Roundworm

المستديرة

Tapeworm

الشريطية



Flukeworm

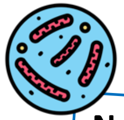
المفلطحة

Helminths Infections

- Multicellular parasites

Types:

1. Nematodes (roundworms, pinworm)
2. Cestodes (tapeworms)
3. Trematodes (flukes)



Names origin is Greek:

- *Nema* = threads. *Odes* = like
- *Cestus* = belt
- *Trema* = hole (refer to sucker pores)

Medication Choice

For helminths , think where **the worm lives** in the body.
(**not** luminal vs tissue)

Intestinal worms	Tissue worms
Stay in gut	Liver, lungs, blood
Albendazole works well	Need drugs that reach tissues. Example: ▪ Echinococcus → liver cysts → Albendazole ▪ Schistosoma → blood vessels → Praziquantel

Albendazole / Mebendazole

Albendazole basically **starves** the parasite to death

MOA:

- Broad-spectrum anti-helminthic (covers most intestinal worms)
- Inhibits microtubule formation → prevent the parasite from absorbing glucose → energy depletion(↓ ATP) → parasite death

Pharmacokinetics:

- Distribute well throughout body tissues (brain, liver, lungs, and muscle)
- Absorption enhanced with a fatty meal

Clinical Use: (Albendazole)

- **Broad-spectrum:**

1. Enterobiasis (pinworm)
2. Ascariasis
3. Hookworm

- **Also used in:**

1. Hydatid disease (a tapeworm disease, forms slow-growing, fluid-filled cysts, in the liver and lungs)

Adverse Effects: (Albendazole)

1. GI upset
2. Elevated liver enzymes
3. Bone marrow suppression (rare)

Helminths Infections (other medications)

Ivermectin

MOA:

- Enhances GABA-mediated chloride influx
- Causes paralysis of parasite

Clinical Use:

Strongyloidiasis (داء الأسطوانيات)

(stron-jih-LOY-deez)

Praziquantel

MOA:

- Increases calcium permeability
- Leads to muscle contraction and paralysis

Clinical Use:

Schistosomiasis (داء البلهارسيا)

(shis-toh-soh-MY-uh-sis)

IMPORTANT

- Peri-anal itching
- Anemia
- Cysts

Pattern Recognition (Helminths)

Clinical Clue	Likely Parasite Type	Key Example	First-line Drug
Peri-anal itching (at night)	Nematode	Enterobius (pinworm)	Albendazole
Iron deficiency **anemia**	Nematode	Hookworm	Albendazole
Visible worm in stool	Nematode	Ascaris	Albendazole
Cysts in liver/lung	Cestode	Echinococcus	Albendazole
Blood in urine (endemic area)	Trematode	Schistosoma	Praziquantel

Enterobiasis (en-tuh-roh-BIGH-uh-siss) **Ascariasis** (as-kuh-RIE-uh-sis) **Echinococcus** (i-KYE-nuh-KAH-kuhs) **Schistosomiasis** (shis-toh-soh-MY-uh-sis)

ECTOPARASITES

الطفيليات الخارجية

Parasites living on skin surface causing itching and inflammation

Scabies Mites
عث الجرب

Lice
قمل

Tick
قرادة

Ectoparasites (Overview)

- Live on skin surface
- Cause itching and inflammation
- Transmission is usually via close contact

Medication Choice

Location → skin surface, so treatment is:

1. Topical first → Permethrin
2. Systemic (only if severe) → Ivermectin

Ticks (clinical differentiation point only)

- Attach to skin and feed on blood
- Managed by mechanical removal (**not antiparasitic drugs**)
- Important due to disease transmission

Treatment

- **Permethrin**

MOA: Disrupts sodium channels → neurotoxicity in parasites

Uses: First-line for scabies and lice

- **Ivermectin**

- Used in severe or resistant cases
- Oral option for widespread infestation

IMPORTANT

Minimal itching
Disease transmission
Tick removal

Pattern Recognition (Ectoparasites)

Feature	**Scabies**	**Lice**	**Ticks**
Location	Inside skin	On hair	Attached to skin
Itching	Severe, at night	Mild-moderate	**Minimal**
Key finding	Burrows	Nits (eggs)	Visible tick
Risk	Skin infestation	Hair infestation	**Disease transmission**
Treatment	Topical Permethrin +/- oral Ivermectin	Permethrin (topical)	Physical removal of the tick + monitoring (symptoms)
Image			

Skin Burrows

Lice Nits

Tick Removal

References

1. Lippincott's Illustrated Reviews: Pharmacology
2. Katzung's Basic and Clinical Pharmacology
3. Goodman & Gilman's The Pharmacological Basis of Therapeutics

FROM ANCIENT GREEK

parasite
/ˈparəsʌɪt/

Para - "besides, alongside"

-sītos "grain, food".

TRANSLATED AS:

"one who eats at another's table"

It referred to a person who habitually ate at someone else's expense !

It became a biological term in the 16th century.